Extracted Content

TABLE I
nFAPI Delay Management Timing Offset Parameters [1]

Parameter	Type	Description
Timing offsets (DL_TTI, UL_TTI, UL_DCI, Tx_Data)	uint32_t	The timing offset (μs) before the slot start that the corresponding request must be received at PHY.

TABLE II
nFAPI Timing Info Parameters [1]

Field	Type	Description
Last SFN / Slot	uint16_t	The completed SFN (0–1023) and slot (0–159) that triggered the message.
Time Since Last Info	uint32_t	Time in ms since the last Timing Info message was sent.
Jitter	uint32_t	Inter-message jitter (μs) for DL_TTI, TxData, UL_TTI, and UL_DCI.
Latest Delay	int32_t	Latest delay offset (μs) from acceptable window. Positive: late, negative: early.
Earliest Arrival	int32_t	Earliest arrival offset (μs) from acceptable window.
Subcarrier Spacing	uint8_t	SCS index (0: 15kHz to 4: 240kHz) as defined in TS38.211.

TABLE III
nFAPI Delay Management Configuration Parameters

Parameter	Type	Description
timingWindow	uint16_t	Duration of the sliding window used for delay calculation.
timingMode	uint8_t	Reporting mode for Timing Info: Periodic, Aperiodic, or Both.
timingPeriod	uint8_t	Transmission interval for Timing Info messages.

$$\Delta t_{offset} = \frac{(t_2 - t_1) - (t_4 - t_3)}{2} \tag{1}$$

$$T_{owd} = \frac{(t_4 - t_1) - (t_3 - t_2)}{2} \tag{2}$$

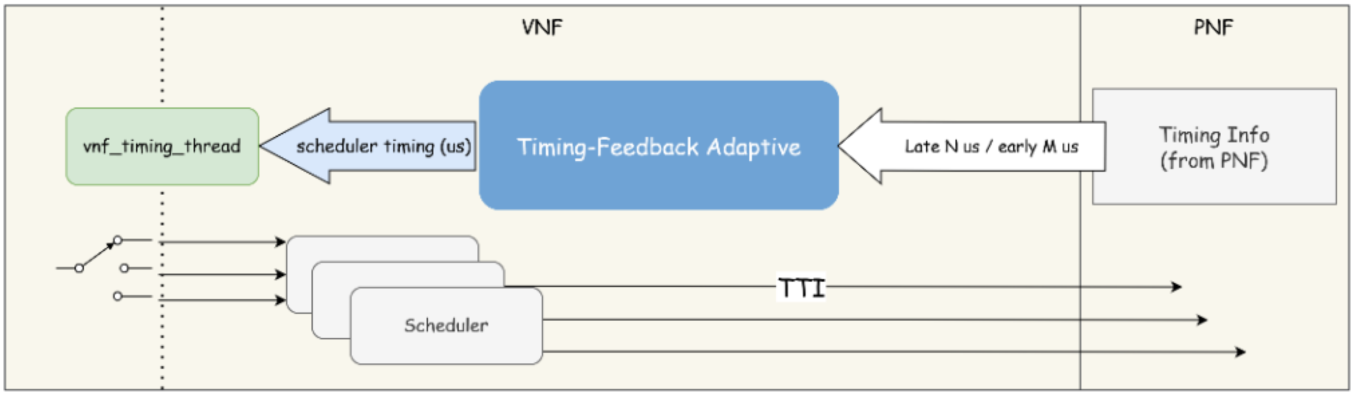

Fig. 1. System Model of the Timing-Feedback Adaptive Scheduler.

$$\Delta t_{total} = \Delta t_{offset} + T_{margin} \tag{3}$$

$$N_{slot} = \lfloor \frac{\Delta t_{total}}{T_{slot}} \rfloor \tag{4}$$

$$T_{us} = \Delta t_{total} \pmod{T_{slot}} \tag{5}$$

$$A_{us} = -T_{us} \tag{6}$$

$$A_{slot} = N_{slot} \tag{7}$$

$$N_{skip} = \lfloor \frac{\Delta t_{accumulated}}{T_{slot}} \rfloor \tag{8}$$

References

[1] Small Cell Forum, “5g nfapi specification,” Small Cell Forum, Tech. Rep., July 2022, table 4-3: Timing Info parameters.